# CSE629.02 | **Deep Learning** | Introduction

Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

**Md. Masum Bhuiyan**
Lecturer, Computer Science and Engineering

It’s Getting Harder to Spot a Deep Fake Video | Bloomberg Originals | 8.1M views | 5 years

**Prompt**: an armchair in the  shape of an avocado. | DALL-E | Open AI

Super resolution | SRGAN

LLM | ChatGPT

**You**

Hello GPT. Could you please tell my students why they should enroll to my course deep learning in short?

**ChatGPT**

Hello! Enrolling in a deep learning course can offer your students numerous benefits:

1. **Cutting-Edge Technology:** Deep learning is at the forefront of artificial intelligence, enabling advancements in areas such as image and speech recognition, natural language processing, and more.
2. **Career Opportunities:** Deep learning skills are in high demand across various industries, including technology, healthcare, finance, and automotive. By learning deep learning, students can enhance their employability and open doors to exciting career opportunities.
3. **Problem Solving:** Deep learning provides powerful tools for solving complex problems, making it a valuable skill for addressing real-world challenges. This course can empower students to develop innovat solutions using state-of-the-art techniques.

## Artificial Intelligence

Any technique that enables computers to mimic human behavior.

## Machine Learning

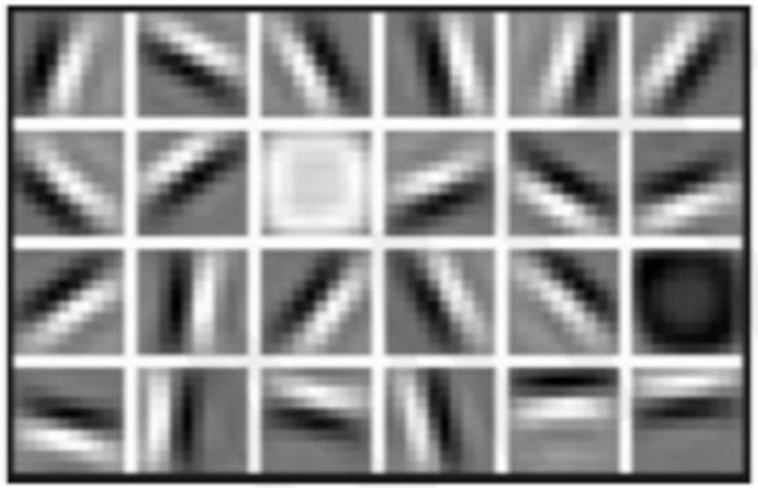
Ability to learn without being explicitly programmed.

## Deep Learning

Extract patterns from data using neural nets

Machine Learning requires hand engineered features
that are time consuming, not scalable in practice.

**Can we learn this underlying features just from data?**



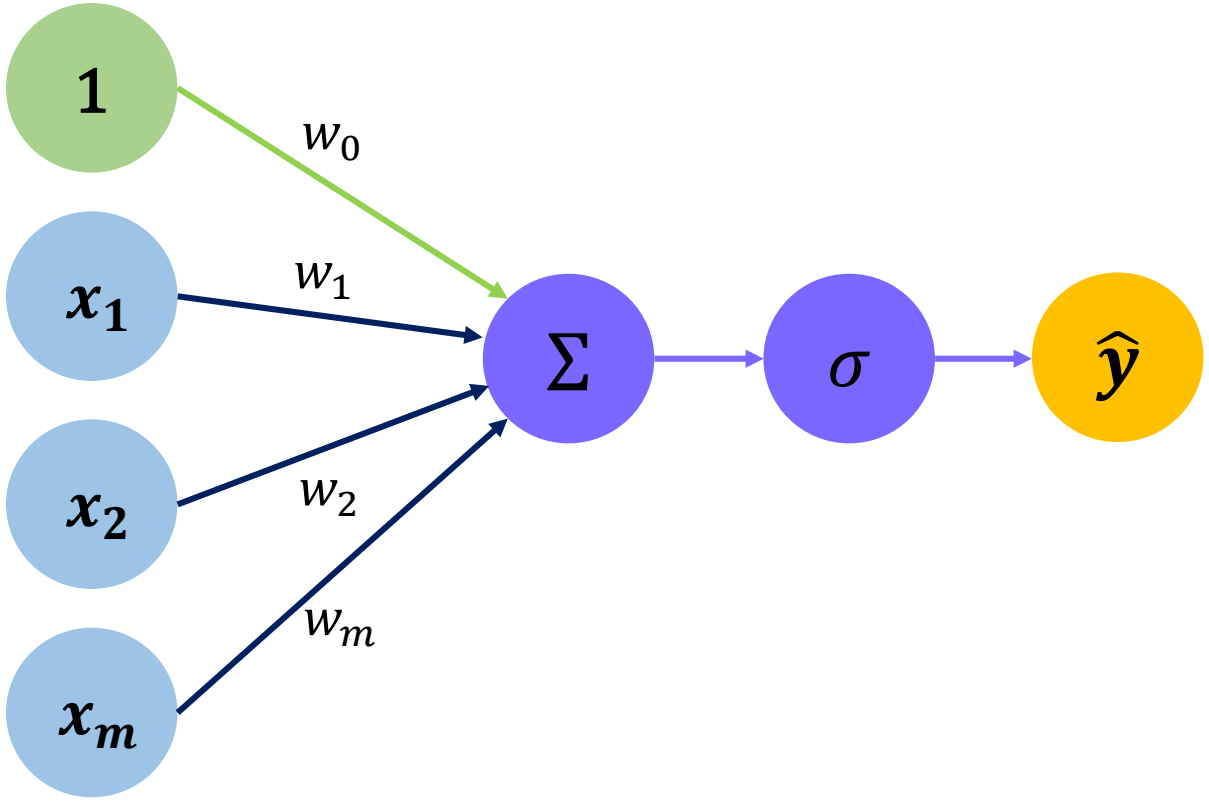
**Low Level Features**
Edges & Lines

**Mid Level Features**
Eyes, Nose & Ears

**High Level Features**
Facial Structure

# The Perceptron

The structural building block of deep learning



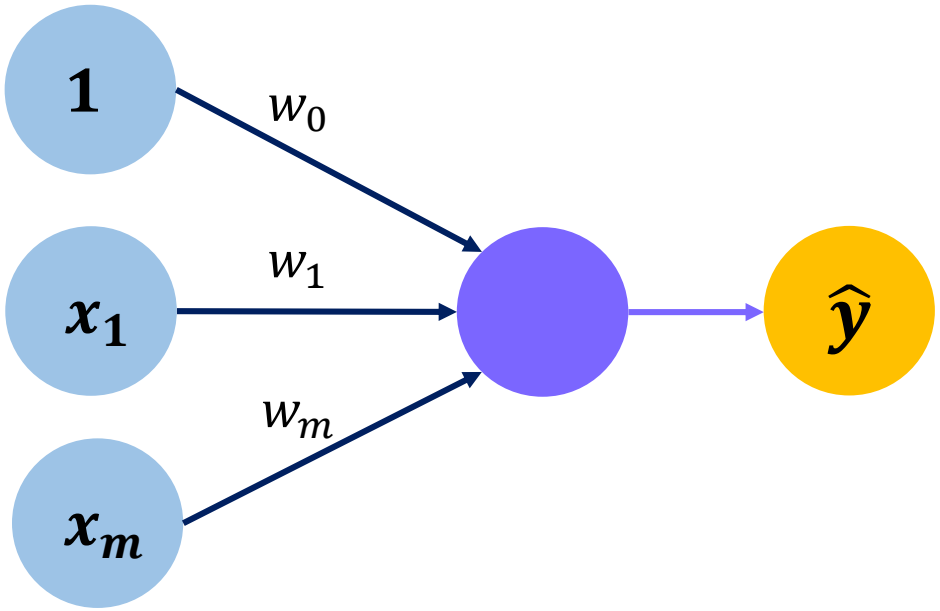

Linear combination of inputs

Output

$$\hat{y} = \sigma\left(w_0 + \sum_{1}^{m} x_i w_i\right)$$

Non linear activation function

Bias

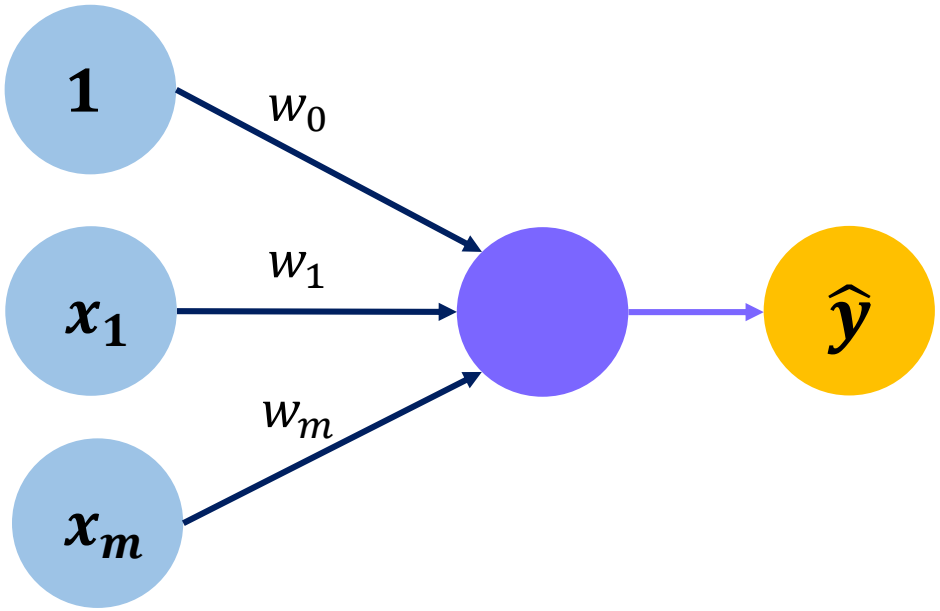



$$\hat{y} = \sigma\left(w_0 + \sum_{1}^{m} x_i w_i\right)$$

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \qquad \boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}$$

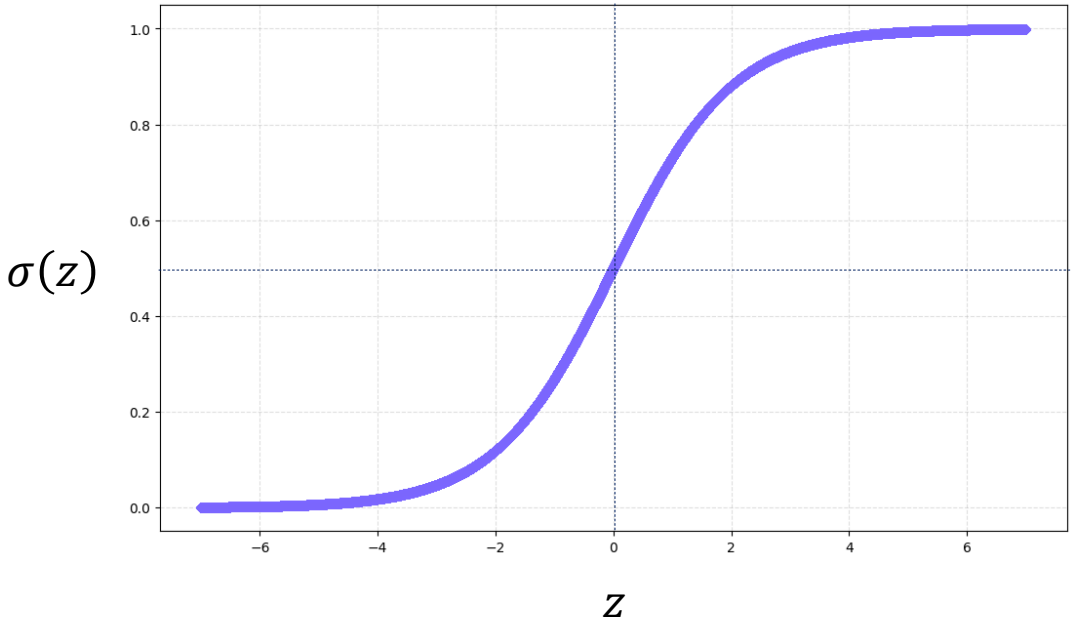
$$z = \boldsymbol{x}^T \boldsymbol{w} + w_0$$

$$\hat{y} = \sigma(z)$$

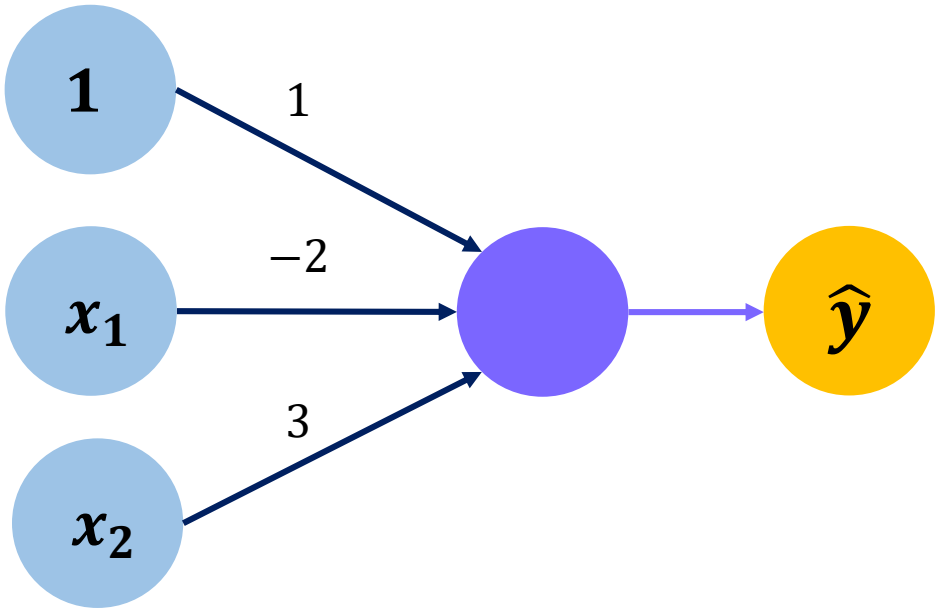



- Example: sigmoid function.

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$




tf.math.sigmoid(z)



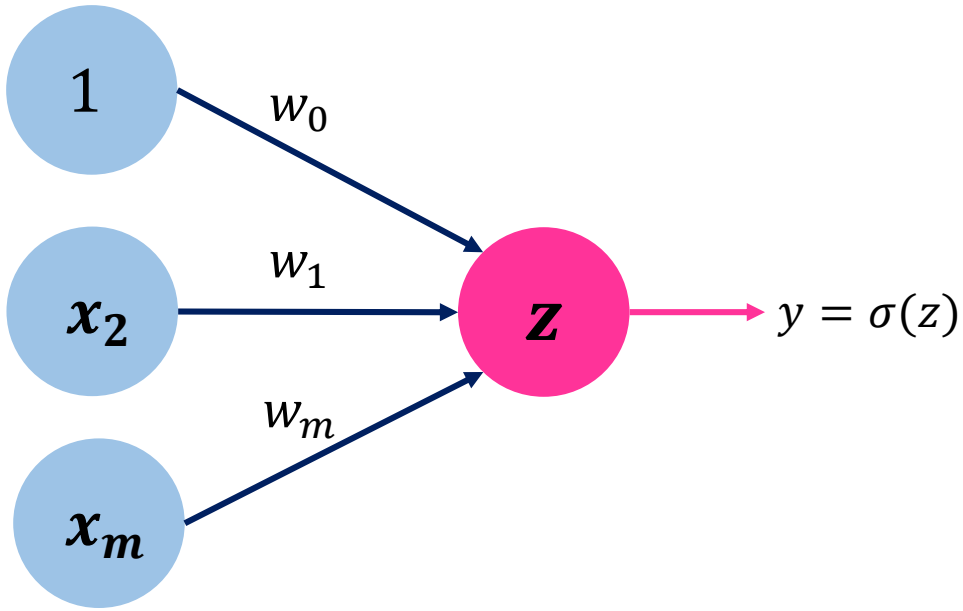

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad \boldsymbol{w} = \begin{bmatrix} -2 \\ 3 \end{bmatrix} \qquad w_0 = 1$$

$$z = 1 - 2x_1 + 3x_2$$

$$\hat{y} = \sigma(1 - 2x_1 + 3x_2)$$

If $\boldsymbol{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$,

$$\begin{aligned} \hat{y} &= \sigma(1 - 2x_1 + 3x_2) \\ &= \sigma(8) \\ &\approx 0.9996 \end{aligned}$$

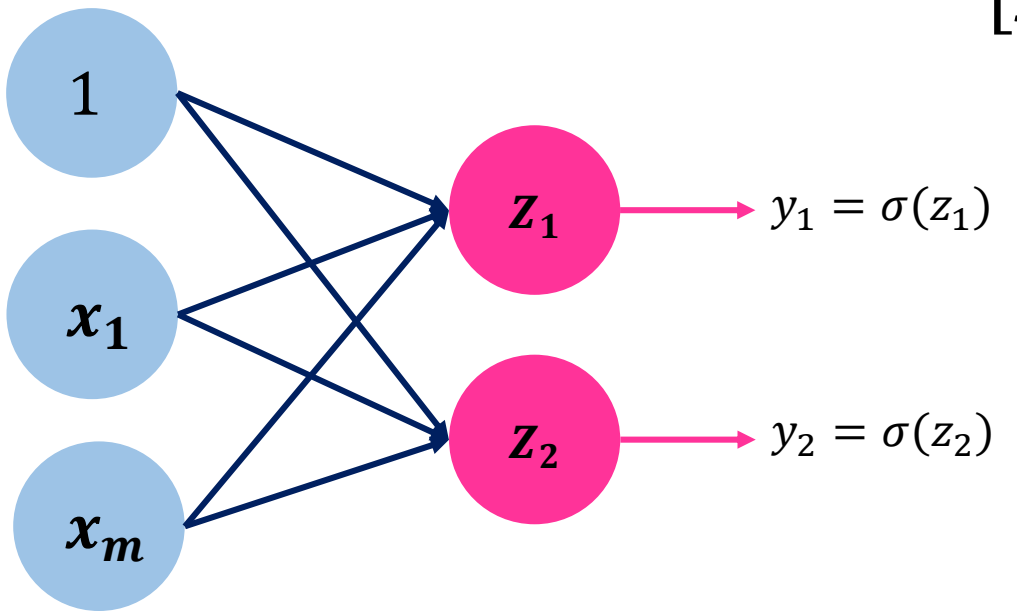



$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \qquad \boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}$$

$$z = \boldsymbol{x}^T \boldsymbol{w} + w_0$$

$$\hat{y} = \sigma(z)$$

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \qquad \boldsymbol{w} = \begin{bmatrix} w_{1,1} & w_{1,2} & & w_{1,n} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,n} \\ \vdots & \vdots & & \vdots \\ w_{m,1} & w_{m,2} & & w_{m,n} \end{bmatrix} \qquad \boldsymbol{w}_0 = \begin{bmatrix} w_{0,1} \\ w_{0,2} \\ \vdots \\ w_{0,n} \end{bmatrix}$$




$$z = \boldsymbol{x}^T \boldsymbol{w} + \boldsymbol{w}_0$$

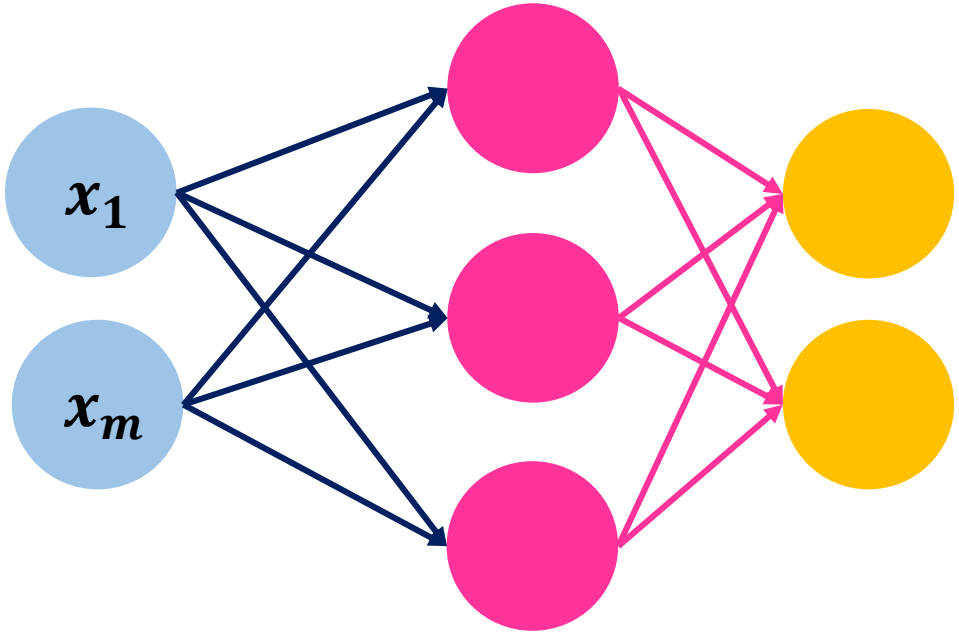
$$\hat{y} = \sigma(z)$$

```python
class MyDenseLayer(tf.keras.layers.Layer):

    def __init__(self, m, n):
        super(MyDenseLayer, self).__init__()
        # Initialize weights and bias
        self.W = self.add_weight([m, n])
        self.b = self.add_weight([1,n])



    def call(self, X):
        # Forward propagates the inputs
        z = tf.matmul(X, self.W) + self.b
        # Feed through a nonlinear activation function
        output = tf.math.sigmoid(z)
        return output
```




```python
import tensorflow as tf

model=tf.keras.Sequential([
    tf.keras.layers.Dense(3),
    tf.keras.layers.Dense(2)
])
```

# Thanks all.

**Next:** CSE629.03 | Deep Learning | **Linear Algebra**